

Module 011 Contradiction

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Abstract—This module forms a critical component of the CEREBRUM framework, a community-structured graph attention engine for Knowledge Graph reasoning. It focuses on Advanced Graph Attention Analysis.

Index Terms—Knowledge Graph, Graph Attention, DSCF, CSA, Hierarchical Reasoning.

1 CONTRADICTION MATERIALIZATION: FACTUAL CONFLICT AS A FIRST-CLASS SIGNAL IN KNOWLEDGE GRAPHS

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1.0.1 Abstract

Autonomous Knowledge Graphs often encounter conflicting information from heterogeneous data sources. Traditional approaches attempt to resolve these conflicts via majority voting or source-trust weighting [?], [?], potentially discarding valuable signals of discovery or debate. We propose **Contradiction Materialization**, a framework that identifies logical and structural inconsistencies and reifies them as queryable `CONTRADICTS` edges. We define five typologies of graph contradiction, including predicate conflict, temporal anachronism, and structural cycle violation. We introduce the **Delta-Authority** metric to quantify the reliability gap between conflicting facts. In v1.2.0, we integrate this engine with the **REM Cycle** skeptical decay protocol, ensuring that false contradictions are pruned while significant intellectual debates are preserved. Our results demonstrate that materializing contradictions improves reasoning robustness by allowing engines to follow exploratory paths across unsettled knowledge boundaries.

1.0.2 1. Introduction

Knowledge representation has long prioritized consistency. However, in scientific research and intelligence analysis, consistency is often an artifact of premature filtering. True intelligence requires the ability to maintain multiple conflicting hypotheses. We demonstrate that reifying conflict as a topological feature—rather than a data quality error—enables more sophisticated multi-hop reasoning.

1.0.3 2. Methodology

1.0.3.1 2.1 Detection Typology: We define a conflict operator $\mathcal{C}(t_1, t_2)$ that evaluates pairs of triples (s, r, o) .

- **Functional Conflict:** $r \in \mathcal{R}_{func} \wedge o_1 \neq o_2$.
- **Temporal Sequence:** $t_{event1} \gg t_{event2}$ where logical order is $1 \rightarrow 2$.
- **Structural Cycle:** Violation of DAG constraints in hierarchical relations.

1.0.3.2 2.2 Reification and Delta-Authority: When $\mathcal{C}(t_1, t_2) = \text{True}$, we materialize edge E_{o_1, o_2} with relation `CONTRADICTS`. The edge is assigned an authority delta ΔA :

$$\Delta A = |\mathcal{T}(\text{source}_1) - \mathcal{T}(\text{source}_2)| \quad (1)$$

where $\mathcal{T}$ is the trust function.

1.0.4 3. Conclusion

Contradiction Materialization transforms Knowledge Graphs from static fact stores into dynamic arenas of evidence. By treating conflict as a structural signal, it provides the necessary foundation for skepticism and dialectical reasoning in autonomous agents.

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